

Client Proposal

1. Entity

The City of Vancouver, Engineering Services is the city department that looks after the roads in Vancouver, British Columbia. It takes care of traffic lights and their timing, road signs, painted lines, turn rules, crosswalks, and the shape and layout of intersections. Vancouver has a busy, mixed road network. Every year the department gets a set amount of money to spend on making roads safer. Every year there are more places that need work than there is money to fix them. So the department has to choose.

2. Problem Definition

We want to fix the intersections where people are most likely to get hurt. That means new or changed signs, different traffic light timing, new crosswalks, turn restrictions, or a redesigned intersection. To choose where, we need to know where people are likely to get hurt before it happens.

Our question is this: for each intersection in Vancouver, named by its pair of cross streets, how many crashes at that intersection will ICBC classify as casualty crashes, meaning at least one person was injured or killed, across the last three complete years of the data?

An intersection with many crashes and no injuries has been lucky, not safe. The same conditions that keep causing collisions there can lead to a serious injury on any given day, so a place that crashes often is a place we should be spending money on, whether or not anyone has been hurt there yet. We want the full crash history of an intersection used as evidence when we predict future injuries, including the crashes where everyone walked away. A high-crash intersection should be able to rise near the top of the ranking on that record alone.

We will then rank every intersection by its predicted number, highest first. The intersections at the top of that ranking are the ones our engineers visit and the ones our budget goes to.

The task

Phase 1: EDA and strategy validation. Using the dataset, examine where and when crashes actually happen in Vancouver.

Phase 2: Data science implementation. We need to know which intersections to fix. Our question is this: for each intersection in Vancouver how many crashes at that intersection will ICBC classify as casualty crashes, meaning at least one person was injured or killed, across the last three complete years of the data?

3. Stakeholder / Audience

The main user is the road safety lead in Engineering Services. This is the engineer who puts together the yearly list of about 25 to 30 locations that get sent forward for a site visit and for funding. That list matters, because a spot has to get a site visit before any sign, signal, or road change there can be paid for.

4. Datasets

5. Success Criteria

A good answer gives us a ranked list of locations, ordered by how many injury crashes we can expect there next year. We care most about the top of the list. If the top 30 spots are right, the work has done its job. Also we need to know what is pushing a location up the list, because that tells our engineer what to look for when they get there.

6. Constraints (Optional)

This data is about crashes only. It says nothing about what signs are already there, how the lights are timed, what the speed limit is, how many lanes there are, or how well lit the street is. It also cannot tell us why any crash happened, only that it did. So in place of a cause, we need you to tell us what the crashes at a high-ranking intersection have in common. That is what our engineer carries with them to the site visit.